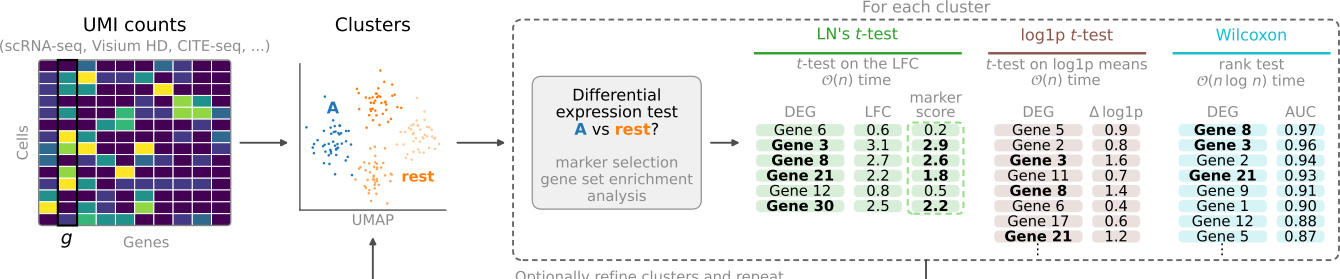
UMI counts
(scRNA-seq, Visium HD, CITE-seq, ...)
Cells
g
Genes
Clusters
A
rest
UMAP
For each cluster
Differential expression test
A vs rest?
marker selection
gene set enrichment analysis
LN's t-test
t-test on the LFC
O(n) time
DEG
LFC
marker score
Gene 6 0.6 0.2
Gene 3 3.1 2.9
Gene 8 2.7 2.6
Gene 21 2.2 1.8
Gene 12 0.8 0.5
Gene 30 2.5 2.2
log1p t-test
t-test on log1p means
O(n) time
DEG
Δ log1p
Gene 5 0.9
Gene 2 0.8
Gene 3 1.6
Gene 11 0.7
Gene 8 1.4
Gene 6 0.4
Gene 17 0.6
Gene 21 1.2
Wilcoxon
rank test
O(n log n) time
DEG
AUC
Gene 8 0.97
Gene 3 0.96
Gene 2 0.94
Gene 21 0.93
Gene 9 0.91
Gene 1 0.90
Gene 12 0.88
Gene 5 0.87
Optionally refine clusters and repeat